Sales & Forecast Performance Dashboard

ETL Logic

The ETL pipeline was developed in Python using the pandas library to transform raw JSON sales and forecast data into a structured dimensional model suitable for analytics and reporting in Microsoft Power BI.

The pipeline follows the standard ETL process:

1. Extract

- Loaded Sales.json and forecast.json using Python JSON processing.
- Converted the raw JSON records into pandas DataFrames for analysis and transformation.

2. Data Exploration & Data Quality Validation

Several data quality checks were performed before modeling:

Sales Dataset

- Checked dataset shape, schema, and sample records.
- Identified a large number of duplicate rows and removed them using drop_duplicates().
- Converted OrderDate from object/string datatype to datetime.
- Detected invalid dates and excluded unparseable rows from the fact table.
- Removed leading/trailing spaces from all string columns.
- Identified that the Color column contained incorrect values identical to the Subcategory column.
- Dropped the unreliable Color column and derived color information from product names instead.

Forecast Dataset

- Checked schema, null values, and duplicates.
- Standardized string columns by trimming whitespace.
- Validated that forecast values and years were stored as numeric datatypes.

Transformation Logic

Date Dimension (dim_date)

A complete calendar table was generated dynamically using the minimum and maximum sales dates.

The dimension includes:

- Year
- Quarter
- Month
- Week
- Day attributes
- Weekend indicator
- Date labels for reporting

A surrogate DateKey was generated using the YYYYMMDD format.

Year Dimension (dim_year)

Created to solve the granularity mismatch between:

- fact_sales → daily grain
- fact_forecast → yearly grain

This allows a single year slicer in Power BI to filter both fact tables consistently.

Product Dimension (dim_product)

Grain:

- One row per ProductKey

Transformations:

- Extracted Color from the last word of Product Name.

- Created ProductFamily by removing the color from product names when a valid color existed.
- Used "Unknown" for products where color could not be identified.

The Brand column was replaced with BrandKey to enforce dimensional relationships.

Customer Dimension (dim_customer)

Grain:

- One row per CustomerKey

Logic:

- Combined customer identity attributes with the best available profile information.
- Since most customer profile fields were missing in the source data, missing values were replaced with "Unknown".
- Standardized customer codes using uppercase formatting and whitespace trimming.
- Geography attributes were linked through dim_geography.

Brand Dimension (dim_brand)

Created as a conformed dimension to support analysis across:

- fact_sales
- fact_forecast

A surrogate BrandKey was generated for relational consistency.

Geography Dimension (dim_geography)

Grain:

- One row per country/region

Contains:

- CountryRegion

- Continent

A surrogate CountryRegionKey was created and later used in both customer and forecast tables.

Fact Sales (fact_sales)

Grain:

- One row per sales transaction

Contains:

- ProductKey
- CustomerKey
- DateKey
- Quantity
- NetPrice
- TotalSale

TotalSale was calculated as:

$$\text{TotalSale} = \text{Quantity} \times \text{NetPrice}$$

A surrogate SaleID was added for uniqueness.

Fact Forecast (fact_forecast)

Grain:

- One row per Year × Brand × CountryRegion

Contains:

- BrandKey
- CountryRegionKey
- Year
- Forecast

A surrogate ForecastID was generated.

Data Model

The final model follows a dimensional star/galaxy schema design.

Fact Tables

- fact_sales
- fact_forecast

Dimension Tables

- dim_date
- dim_year
- dim_product
- dim_customer
- dim_brand
- dim_geography

Relationships

Sales Model

- fact_sales[DateKey] → dim_date[DateKey]
- fact_sales[ProductKey] → dim_product[ProductKey]
- fact_sales[CustomerKey] → dim_customer[CustomerKey]

Forecast Model

- fact_forecast[BrandKey] → dim_brand[BrandKey]
- fact_forecast[CountryRegionKey] → dim_geography[CountryRegionKey]
- fact_forecast[Year] → dim_year[Year]

Shared Dimensions

- dim_brand connects forecast and product analysis.

- dim_year bridges yearly forecast data with daily sales data.

All relationships in Power BI were configured as:

- One-to-many
- Single cross-filter direction

Key Assumptions

1. Product color appears as the last word in the product name when available.
2. Product names without recognizable colors are assigned "Unknown" color.
3. Duplicate rows in the sales dataset were considered invalid and removed.
4. Invalid or unparseable dates were excluded from the fact table.
5. Customer profile fields (Name, Education, Occupation) were incomplete in the source system; missing values were replaced with "Unknown".
6. Forecast data is stored at yearly granularity, while sales data is stored at daily transaction granularity.
7. Brand and CountryRegion were modeled as shared dimensions to support cross-analysis between sales and forecast datasets.

Final Output

The transformed datasets were exported as structured CSV files representing:

- Dimension tables
- Fact tables